

R3 SCRUB SERVICE MANUAL

1. INTRODUCTION

This manual describes the operation and routine maintenance of the machine. It is intended for operators who have been authorised and trained by the service centre. Read the safety chapter before performing any operation. The machine must be isolated from the supply before any maintenance is carried out, and the format of the safety labels must not be altered. There is a notice on the surface of the control panel.

2. TECHNICAL SPECIFICATION

The rated supply is 240 Vca and the maximum absorbed current is 8.5 A. The protection class is IPX0 and the machine must not be operated outdoors. The sound pressure level is 68 dB and the total mass with batteries is 143 kg.

6. TROUBLESHOOTING

This chapter lists the most frequent faults, their possible causes and the corrective actions that the operator is authorised to perform. If the fault persists after the corrective action has been applied, contact the authorised service centre and provide the fault code shown on the display together with the machine serial number.

PROBLEM	POSSIBLE CAUSE	SOLUTION
THE ROBOT DOES NOT START	The battery is out of charge and the onboard charger is disconnected from robot. the mains	Perform a complete recharge cycle and attempt restarting the
THE ROBOT STOPS SUDDENLY	The E-Stop is triggered, indicated by the red lamp on the control panel	Release the E-stop and press the reset button located on the rear panel.
THE BRUSH DOES NOT ROTATE	The brush deck is not mounted correctly on the chassis frame of the engaged machine	Verify that the deck is mounted and that the locking pin is
THE MACHINE LEAVES STREAKS	The squeegee blade is worn or the recovery tank is full to the maximum level	Replace the squeegee blade and empty the recovery tank before restarting.
NO WATER ON THE FLOOR	The solution filter is clogged or the solution tank is empty after a long cycle	Clean the solution filter and refill the tank with clean water.

6.1 FAULT CODES

The machine reports faults using structured codes shown on the operator display.

CODE	DESCRIPTION	ACTION
AF-01-3021-6-1	Slope is too steep	Push robot away from slope and try again
AF-01-3022-6-1	Slope is too steep	Push robot away from slope and try again
AF-01-3023-6-1	Slope angle sensor fault	Restart the machine and recalibrate the sensor
AE-02-3605-2-4	Brush motor overcurrent	Inspect the brush deck for obstructions
AE-02-3606-2-4	Brush motor overtemperature	Allow the motor to cool before restarting
AE-03-1204-1-2	Solution pump failure	Check the pump connector and the solution filter
AE-03-1205-1-2	Solution level low	Refill the solution tank with clean water
AH-04-2200-3-1	Battery voltage below minimum	Recharge the battery for at least 8 h
AH-04-2201-3-1	Battery temperature high	Stop charging and let the battery cool
AH-05-7710-5-3	Drive wheel encoder mismatch	Inspect the encoder wiring on both wheels
AK-06-4402-2-2	Recovery tank full	Empty the recovery tank and restart the cycle
AK-06-4403-2-2	Recovery tank sensor fault	Clean the float sensor inside the tank
AM-07-9001-4-1	Navigation map missing	Reload the cleaning map from the operator panel
AM-07-9002-4-1	Localisation lost	Move the robot to a known start point and restart

7. ROUTINE MAINTENANCE

Check the squeegee blades daily and replace them when the edge is rounded. Clean the solution filter weekly and inspect the brush deck for entangled debris. The battery connector must be inspected monthly for corrosion and the terminals tightened to 8 Nm.

7.1 SQUEEGEE ASSEMBLY

Remove the squeegee by loosening the two knobs on the support arm. Wash the blades in clean water and inspect the

rubber edge for cuts. A rounded edge leaves streaks on the floor and must be replaced. Refit the assembly and tighten the knobs to 4 Nm.

7.2 BRUSH DECK

Lower the deck fully before removing the brushes. Rotate each brush anticlockwise to release it from the hub. Remove entangled fibres and check that the hub is not worn. The deck must be mounted squarely on the chassis before the machine is restarted.

7.3 SOLUTION AND RECOVERY TANKS

Drain the recovery tank through the hose at the rear of the machine. Rinse both tanks with clean water at the end of every shift. Inspect the float sensor inside the recovery tank and clear any deposits, otherwise the tank-full fault will be reported early.

7.4 BATTERY AND CHARGER

The battery must be charged in a ventilated area. Do not interrupt a charge cycle before completion. Inspect the connector monthly for corrosion and tighten the terminals to 8 Nm. A battery held below the minimum voltage for a long period will not recover.

7.5 DRIVE SYSTEM

Check the drive wheels for embedded debris and confirm that both encoders report the same distance after a test run. A mismatch between the two encoders indicates a wiring fault and is reported by the machine as a drive system fault code.

8. SAFETY

Only personnel authorised and trained by the service centre may open the electrical compartment. Isolate the machine from the supply and wait five minutes before removing any cover. The machine must not be washed with a pressure washer. Do not operate the machine on a slope steeper than the maximum stated in the technical specification.

9. SPARE PARTS

Order spare parts using the part number shown below together with the machine serial number. Parts marked with a rating must be replaced with an identical rating.

PART NUMBER	DESCRIPTION	QTY	RATING
AE-02-3605-2-4	Brush motor assembly 24 V	1	8.5 A
AE-02-3610-2-4	Brush hub, left	1	-
AE-02-3611-2-4	Brush hub, right	1	-
AK-06-4402-2-2	Recovery tank float sensor	1	24 Vcc
AK-06-4410-2-2	Recovery tank drain hose	1	-
AH-04-2200-3-1	Battery pack 240 Vca charger	1	8.5 A
AH-04-2210-3-1	Battery connector kit	1	-
AM-07-9001-4-1	Operator panel display	1	24 Vcc
AM-07-9010-4-1	Emergency stop button	1	-
AF-01-3021-6-1	Slope sensor module	1	24 Vcc
AS-08-5501-1-1	Squeegee blade, front	2	-
AS-08-5502-1-1	Squeegee blade, rear	2	-
AS-08-5510-1-1	Squeegee support arm	1	-
AP-09-6600-2-3	Solution pump 24 V	1	3.2 A
AP-09-6610-2-3	Solution filter cartridge	1	-
AD-10-7700-5-3	Drive wheel encoder	2	5 Vcc